Rithwak Somepalli

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Education

Georgia State University — Atlanta, GA Expected May 2027

B.S. in Computer Science | **Coursework:** Data Structures, Artificial Intelligence, Machine Learning, Software Engineering, Database Systems

Experience

Software Development Engineering Intern | Twin Brain, Evu — Remote Jun. 2025 – Aug. 2025

- Built backend services integrating AI trading systems with external broker APIs, handling high-throughput simulation workloads during live market hours.
- Implemented real-time health telemetry dashboards and automated anomaly alerts, reducing incident response time during simulation runs.
- Verified data pipeline outputs and model predictions using internal reconciliation tooling to ensure consistency across trading environments.

Projects

Cloud-Native Game Server Orchestrator | Terraform, PowerShell, Bash, Python, Flask Oct. – Nov. 2025

- Built a one-click deployment pipeline using PowerShell and Terraform to automate provisioning of OCI compute, VCN networking, security lists, and firewall rules.
- Wrote a Bash cloud-init template configuring iptables and a systemd-managed Flask REST API for remote start/stop/status management of the game server.
- Automated SSH key generation via Terraform's tls_private_key provider and used file provisioners to upload game modifications to the server at deploy time.

Heart Anomaly Detection System | PyTorch, FastAPI, Docker, Kubernetes Dec. 2025

- Designed a decreasing-width early-exit neural architecture in PyTorch with learnable confidence thresholds and a compound loss balancing accuracy, efficiency, and carbon cost.
- Containerized with Docker, authored Kubernetes manifests (Deployments, HPA, liveness probes, PVC), and streamed real-time predictions via FastAPI and WebSockets.
- Used a dual learning-rate optimizer strategy, training model weights and learnable exit thresholds at separate rates to prevent routing collapse during early training.

Co-Evolutionary Market Simulation | TensorFlow, Keras, NumPy, Pandas Aug. – Oct. 2025

- Built a bid-ask market simulation where a Double DQN agent quotes spreads against an EGT population evolving via replicator dynamics over real OHLCV data.
- Implemented a hybrid reward signal combining PnL, normalized returns, inventory risk penalties, and mean-reversion incentives to enforce market-neutral behavior.
- Computed post-simulation performance analytics including annualized Sharpe ratio, max drawdown, and return distribution skew/kurtosis to evaluate strategy robustness.

Distributed Network Intrusion Detection System | Node.js, Express, Linux, OCI Sept. 2025

- Built a honeypot-based NIDS in Node.js, deploying traps on HTTP, FTP, SSH, and UDP ports with regex detection for SQLi, XSS, directory traversal, brute force, and RCE.
- Streamed classified threat events to a centralized OCI-hosted Express dashboard for real-time remote monitoring from any browser.
- Designed protocol-specific traps for HTTP, FTP, SSH, and UDP, each parsing different payloads (URLs, credentials, handshake banners, raw datagrams) to maximize detection coverage.

Technical Skills

Languages: Python, JavaScript, Bash, SQL, PowerShell, Java, C++, HCL
Frameworks: PyTorch, TensorFlow, Keras, FastAPI, Flask, Node.js, Express, NumPy, Pandas
Cloud & Infra: Terraform, Kubernetes, Docker, Oracle Cloud (OCI), GitHub Actions, Linux
Certifications: Google Cybersecurity Certificate — May 2023